Week 2: Linear Regression*

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1 Warm-Up: Linear Regression in 1-d

Consider the following task: you are given n pairs of numbers $(x_1, y_1), \dots, (x_n, y_n)$, and the goal is to find the line $y = mx + q$ that minimizes the (empirical) mean square error:

$$\hat{R}(m, q) = \frac{1}{n} \sum_{i=1}^{n} (mx_i + q - y_i)^2.$$

This is a polynomial in m and q, goes to infinity as m and q grow in absolute value, and is thus minimized in the critical value of the gradient. We set to zero the derivative with respect to q:

$$\frac{\partial \hat{R}}{\partial q} = \frac{2}{n} \sum_{i=1}^{n} (mx_i + q - y_i) = 0 \implies q = \frac{1}{n} \sum_{i=1}^{n} (y_i - mx_i) = \overline{y} - m\overline{x},$$

where $\overline{x} = \sum_i x_i/n$ and $\overline{y} = \sum_i y_i/n$. We move our attention to m, and compute the corresponding derivative:

$$\frac{\partial \hat{R}}{\partial m} = \frac{2}{n} \sum_{i=1}^{n} x_i(mx_i + q - y_i) = 0.$$

By plugging the value $q = \overline{y} - m\overline{x}$ in the last equality, we get:

$$\sum_{i=1}^{n} x_i(mx_i - m\overline{x} + \overline{y} - y_i) = 0 \implies m = \frac{\sum_{i=1}^{n} x_i(y_i - \overline{y})}{\sum_{i=1} x_i(x_i - \overline{x})}.$$

While the above formula is complete, there is a cleaner version of it that is usually reported. It can be obtained by massaging the numerator and the denominator:

$$\frac{1}{n} \sum_{i=1}^{n} x_i(y_i - \overline{y}) = \frac{1}{n} \sum_{i=1}^{n} (x_i y_i - \overline{y} x_i \pm y_i \overline{x}) = \frac{1}{n} \sum_{i=1}^{n} (x_i y_i - \overline{y} x_i - y_i \overline{x} + \overline{x}\overline{y}) = \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})$$

$$\frac{1}{n} \sum_{i=1}^{n} x_i(x_i - \overline{x}) = \frac{1}{n} \sum_{i=1}^{n} \left(x_i^2 - \overline{x} x_i \pm x_i \overline{x}\right) = \frac{1}{n} \sum_{i=1}^{n} \left(x_i^2 - 2\overline{x} x_i + \overline{x}^2\right) = \sum_{i=1}^{n} (x_i - \overline{x})^2$$

All in all, we have the classical formula for the one-dimensional *ordinary least-squares estimators*:

$$\hat{m} = \frac{\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n} (x_i - \overline{x})^2}, \quad \hat{q} = \overline{y} - \hat{m}\overline{x}.$$

*These notes are meant as a teaching support for the students of the "Machine Learning" course for the Master's Degree in Engineering in Computer Science and Artificial Intelligence that I am teaching in Fall 2025. They are likely full of typos and imprecisions, so please get in touch with me in class if something feels off. From time to time I may update new versions of these notes, so make sure to download the lastest version.

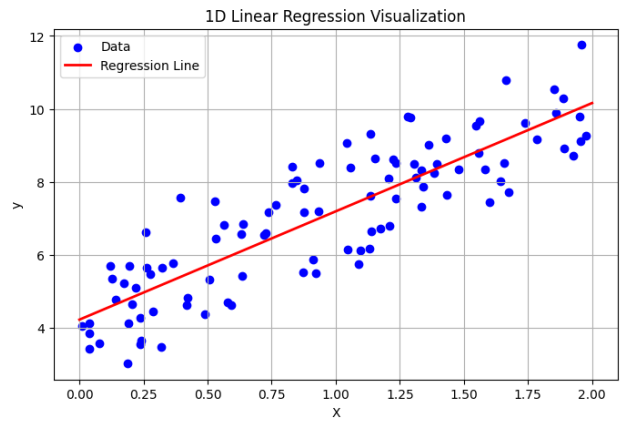

Figure 1: Visualization of linear regression in 1d.

2 The Linear Regression Framework

In the one-dimensional setting, we found the linear function (the line) that minimizes the mean squared error. We can generalize this approach in two directions: (i) considering higher dimensions, (ii) considering functions that are linear in the parameters, and not on the points.

We are given n points $(x_1, y_1), \ldots, (x_n, y_n)$, where each $x_i \in \mathcal{X}$, and $y_i \in \mathbb{R}$. On the space $\mathcal{X}$, we are given d feature maps $\phi_i : \mathcal{X} \to \mathbb{R}$, which induce functions of the type $f_\theta : \mathcal{X} \to \mathbb{R}$ for $\theta \in \mathbb{R}^d$ as follows:

$$f_\theta(x) = \sum_{j=1}^{d} \phi_j(x) \cdot \theta_j = \phi(x)^T \theta, \tag{1}$$

where $\phi(x)$ is the d-dimensional vector whose j^{th} entry is $\phi_j(x)$. Note, this problem is still called *linear* regression because function $f_\theta(x)$ is not linear in x, but is linear in θ!

Our optimization problem. Our goal is to find $\theta \in \mathbb{R}^d$ that minimizes the mean squared error on the input points with respect to the feature maps $\phi_1, \ldots, \phi_d$. Stated differently, we want to find the vector θ, that minimizes:

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^{n} (y_i - f_\theta(x_i))^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - \phi(x)^T \theta)^2 = \frac{1}{n} ||y - \Phi\theta||_2^2.$$

The last version of the mean squared error is more compact as it uses vector and matrix notation. But don't be scared: y is the n-dimensional vector whose i^{th} component is simply y_i: the y value of the ith input point. Similarly, Φ is the $n \times d$ matrix obtained by stacking one onto the others the vectors $\phi(x_1)^T, \ldots \phi(x_n)^T$; in particularly, the generic $\Phi_{i,j}$ element is $\phi_j(x_i)$. Note, the matrix Φ does depend on the x_i.

Example 2.1 (1-d Linear Regression)**.** *If we get back to the one-dimensional example, we see that it is camputered by Equation* (1)*, by setting* $\phi_0 = 1$*, and* $\phi_1 = x$*.*

Example 2.2 (Polynomial Regression)**.** *We can generalize the above example and consider polynomial regression so that* $f_\theta(x) = \sum_{j=0}^{\ell} \theta_j \phi_k(x)$*, where* $\phi_k = x^k$*. In general, we can capture* ℓ*-degree polynomials with* $\ell + 1$ *dimensional* θ*.*

*We start from ϕ_0 and not ϕ_1 to stress that the generic term ϕ_k corresponds to x^k.

Example 2.3 (Fourier Regression)**.** *If the function we are trying to estimate look periodic, it may be a good idea to consider as feature maps functions like* $\sin(\frac{2\pi x}{T}i)$ *and* $\cos(\frac{2\pi x}{T}i)$*, for varying* i*.*

2.1 Ordinary Least Squares Estimator

It turns out that there is a simple expression for the regression problem, which is called the Ordinary-Least Squares Estimator (OLS):

Theorem 2.4 (OLS Estimator)**.** *Let* $(x_1, y_1), \ldots, (x_n, y_n)$ *be points in* $\mathcal{X} \times \mathbb{R}$*, and* $\phi_1, \ldots, \phi_d$ *be functions from* $\mathcal{X} \to \mathbb{R}$*. If* $\Phi^T\Phi$ *is invertible, then the Ordinary-Least Squares Estimators*

$$\hat{\theta} = (\Phi^T\Phi)^{-1}\Phi^T y \tag{2}$$

is a solution to $\min_{\theta\in\mathbb{R}^d} \hat{R}(\theta)$*.*

Proof. Our goal is to minimize the function $\hat{R}(\theta) = \frac{1}{n}||y - \Phi\theta||_2^2$, which is a degree-two polynomial in the coordinates of θ. Moreover it is coercive, meaning that for $||\theta||_2 \to \infty$, the value of $\hat{R}$ diverges to $+\infty$. This implies that we need to find the value(s) for which the gradient of $\hat{R}$ is zero. To compute this gradient, let's rewrite more explicitely $\hat{R}$:

$$\begin{aligned}
\hat{R}(\theta) &= \frac{1}{n}(y - \Phi\theta)^T(y - \Phi\theta) && \text{(For any vector } v \text{ it holds } ||v||_2^2 = v^Tv) \\
&= \frac{1}{n}\left(y^Ty - (\Phi\theta)^Ty - y^T\Phi\theta - (\Phi\theta)^T(\Phi\theta)\right) && ((A+B)^T = A^T + B^T, \text{ for any matrices } A, B) \\
&= \frac{1}{n}\left(||y||_2^2 - \theta^T\Phi^Ty - y^T\Phi\theta + ||\Phi\theta||_2^2\right) && ((AB)^T = B^TA^T, \text{ for any matrices } A, B) \\
&= \frac{1}{n}\left(||y||_2^2 - 2\theta^T\Phi^Ty + ||\Phi\theta||_2^2\right). && (3)
\end{aligned}$$

The last inequality holds because $\theta^T\Phi^Ty = y^T\Phi\theta$. Indeed, by making the explicit calculations it is easy to see that both $\theta^T\Phi^Ty$ and $y^T\Phi\theta$ are equal to $\sum_{i=1}^n\sum_{j=1}^n \theta_i\Phi_{i,j}y_j$. Starting from Equation (3), we can easily compute the gradient of $\hat{R}$ with respect to θ:

$$\nabla\hat{R}(\theta) = \frac{2}{n}(\Phi^T\Phi\theta - \Phi^Ty).$$

Finally, the statement of the theorem holds by setting the gradient to zero, which is equivalent to solving the following equation:

$$\Phi^T\Phi\theta = \Phi^Ty. \tag{4}$$

Under the assumption that $\Phi^T\Phi$ is invertible, the system admits a unique solution of the form $\hat{\theta} = (\Phi^T\Phi)^{-1}\Phi^Ty$. If the gradient has a unique zero and the function is coercive, this means that $\hat{\theta}$ minimizes $\hat{R}$! □

Note

A necessary condition for $\Phi^T\Phi$ to be invertible is that $d \leq n$. If this is not the case, infact, then the kernel of Φ would be non-empty.

2.2 A Geometric Interpretation

There is a nice geometric way of interpreting what the OLS estimator does, in terms of orthogonal projections on suitable subspaces. We start by recalling a fact from linear algebra.

Consider any (finite-dimensional) vector space V, then for any linear subspace $S \subseteq V$, we can define its orthogonal $S^\perp$ as the linear subspace of all the vectors $v \in V$ *orthogonal* to S (a vector v is orthogonal to S if $v^T s = 0$ for all $s \in S$). We can write $V = S \oplus S^\perp$, meaning that any $v \in V$ can be written *in a unique way* as the sum of some $v_\parallel \in S$ and $v_\perp \in S^\perp$. The vector $v_\parallel$ is called *projection* of v onto S.

Proposition 2.5 (OLS as Projection)**.** *The linear function $\Phi(\Phi^T\Phi)^{-1}\Phi^T$ from $\mathbb{R}^n$ to $\mathbb{R}^n$ is the linear projection onto* $\mathrm{im}(\Phi)$*, the linear subspace generated by the columns of Φ.*

Proof. We prove this result in three steps: first, we argue that for any vector $v \in \mathrm{im}(\Phi)$, it holds that $\Phi(\Phi^T\Phi)^{-1}\Phi^T v = v$, second, we consider any $v \in \mathrm{im}(\Phi)^\perp$, and argue that belongs to the kernel of $\Phi(\Phi^T\Phi)^{-1}\Phi^T$. Finally, we prove the statement of the Proposition.

We start with a generic $v \in \mathrm{im}(\Phi)$. By definition, vector v is spanned by the column vector of Φ, i.e., $v = \sum_{j=1}^d \lambda_j \Phi^{(j)}$ for some coefficient λ_j (and $\Phi^{(j)}$ stands for the jth column vector of Φ). In matricial form, we can write $v = \Phi\lambda$, where $\lambda \in \mathbb{R}^d$. We have the following:

$$\Phi(\Phi^T\Phi)^{-1}\Phi^T v = \Phi(\Phi^T\Phi)^{-1}\Phi^T(\Phi\lambda) = \Phi(\Phi^T\Phi)^{-1}(\Phi^T\Phi)\lambda = \Phi\lambda = v.$$

Moving to the second point in our proof, consider the generic vector v that is orthogonal to $\mathrm{im}(\Phi)$. This means that v is orthogonal to the columns vectors of Φ, which are the *row vectors* of Φ^T. Therefore, $\Phi^T v = 0$. Recall, the zero vector of a linear space always belongs to the kernel of any linear function; therefore, we have the following:

$$\Phi(\Phi^T\Phi)^{-1}\Phi^T v = \Phi(\Phi^T\Phi)^{-1}0 = 0.$$

We have arrived at the last step of the proof: consider the decomposition of a generic v into $v_\parallel$ and $v_\perp$, with respect to the linear subspace $\mathrm{im}(\Phi)$:

$$\begin{aligned}
\Phi(\Phi^T\Phi)^{-1}\Phi^T v &= (\Phi(\Phi^T\Phi)^{-1}\Phi^T)(v_\perp + v_\parallel) \quad (v = v_\perp + v_\parallel \text{ where } v_\parallel \in \mathrm{im}(\Phi), \text{ and } v_\perp \in \mathrm{im}(\Phi)^T) \\
&= (\Phi(\Phi^T\Phi)^{-1}\Phi^T)v_\perp + (\Phi(\Phi^T\Phi)^{-1}\Phi^T)v_\parallel \qquad \text{(By linearity)} \\
&= v_\parallel,
\end{aligned}$$

where the last equality follows from the first two points of the proof. □

This Proposition provides an *algorithmic* and *geometric* insight into the OLS estimator formula:

- First we compute the projection of the y vector onto the d-dimensional linear subspace $\mathrm{im}(\Phi)$
- Then, we find $\hat{\theta}$ that are the *coordinates* of the projected vector $\Phi(\Phi^T\Phi)^{-1}\Phi^T y$ with respect to the basis of the linear subspace provided by the column vectors of Φ. In formula, this corresponds to solving the linear system $\Phi(\Phi^T\Phi)^{-1}\Phi^T y = \Phi^T\theta$.

3 The Linear Assumption and Linear Regression

Last week, we introduced the framework of decision theory to assess and quantify the quality of a predictor. We now want to look at linear regression from this perspective.

Definition 3.1. *We say that a distribution $\mathcal{D}$ over $\mathcal{X} \times \mathbb{R}$ respects the linear assumption if there exists a vector $\theta^\star \in \mathbb{R}^d$, and d-feature maps $\phi_1, \ldots, \phi_d$ such that:*

- $Y = \phi(X)^T\theta^\star + \varepsilon$
- *ε is a random variable independent from X such that $\mathbb{E}[\varepsilon] = 0$, and $\mathbb{E}\left[\varepsilon^2\right] = \sigma^2$.*

In words, the linear assumption imposes no requirement on $X \in \mathcal{X}$, but strictly constraints the value of Y given X: Y is deterministically specified by $\phi(X)^T\theta^\star$, plus an independent and unbiased noise ε. In particular, if the variance term σ^2 goes to zero, then the random points (X, Y) will lie closer and closer to the curve $y = \phi(x)^T\theta^\star$.

Note

From a geometrical perspective, the curve characterized by $y = \phi(x)^T\theta^*$ *is not* a linear subspace in the x variable, but only in the $\phi(x)$ variable.

Under the linear assumption, and a suitable loss, the optimal predictor has a natural structure.

Proposition 3.2 (Optimal Predictor)**.** *Under the linear assumption and the squared loss, the optimal predictor is $\mathbb{E}[Y|X = x] = \phi^T(x)\theta^\star$.*

Proof. We showed last week that the best predictor under the squared loss is $\mathbb{E}[Y|X = x]$, so we just need to compute explicitly that conditional expectation. Remember, we are under the linear assumption, therefore conditioning on the random variable X having value x we know that $y = \phi^T(x)\theta^\star + \varepsilon$, where the only randomness is given by the noise term ε. For any possible realization x of X, we have the following:

$$\mathbb{E}[Y|X = x] = \mathbb{E}\left[\phi(x)^T\theta^\star + \varepsilon|X = x\right] = \phi(x)^T\theta^\star + \mathbb{E}[\varepsilon] = \phi(x)^T\theta^\star.$$

Note, in the second to last equality we have used that ε is independent from X, while the last equality descends from the linear assumption, namely that $\mathbb{E}[\varepsilon] = 0$. □

Important

The last Proposition tells us that in order to minimize the expected risk —under the linear assumption— it is enough to know the vector θ^*. Clearly, such vector is unknwon, so *we need to estimate it from data.*

3.1 Estimating $\theta^\star$

We prove that OLS estimator does estimate correctly the parameter it is trying to learn: θ^*.

Proposition 3.3. *Consider the estimator $\hat{\theta}$ computed on n i.i.d. samples from a distribution $\mathcal{D}$ on $\mathcal{X} \times \mathbb{R}$ that satisfies the linear assumption, then:*

- *The estimator is unbiased: $\mathbb{E}[\hat{\theta}] = \theta^\star$*
- *The variance of the estimator is $\mathbb{V}[\hat{\theta}] = \frac{\sigma^2}{n}\mathbb{E}[\hat{\Sigma}]$,*

where $\hat{\Sigma} = \frac{1}{n}(\Phi^T\Phi)$

Proof. Fix any realization of the x values of the n i.i.d. samples, i.e., $x_1, \dots, x_n$. We denote with $\mathbf{Y}$ the random vector containing the y values of the samples, and with $\mathbf{y}$ a realization of it. Similarly $\boldsymbol{\varepsilon}$ is the random vector whose entries correspond to the n i.i.d. realizations of the random noise ε. We have the following:

$$\begin{aligned}
\mathbb{E}\left[\hat{\theta} | X_1 = x_1, \dots, X_n = x_n\right] &= \mathbb{E}\left[(\Phi^T\Phi)^{-1}\Phi^T\mathbf{Y} | X_1 = x_1, \dots, X_n = x_n\right] \\
&= (\Phi^T\Phi)^{-1}\Phi^T\mathbb{E}\left[\mathbf{Y} | X_1 = x_1, \dots, X_n = x_n\right] \quad (\Phi \text{ only depends on } x) \\
&= (\Phi^T\Phi)^{-1}\Phi^T\mathbb{E}\left[\Phi\theta^\star + \boldsymbol{\varepsilon} | X_1 = x_1, \dots, X_n = x_n\right] \\
&= (\Phi^T\Phi)^{-1}(\Phi^T\Phi)\theta^\star + (\Phi^T\Phi)^{-1}\Phi^T\mathbb{E}\left[\boldsymbol{\varepsilon}\right] \\
&= \theta^\star,
\end{aligned}$$

where in the last equality we have canceled the noise terms as they are all unbiased. We move our attention to the variance and study the difference between $\hat{\theta}$ and $\theta^\star$:

$$\hat{\theta} - \theta^\star = (\Phi^T\Phi)^{-1}\Phi^T(\Phi\theta^\star + \boldsymbol{\varepsilon}) - \theta^\star = (\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon}. \tag{5}$$

The variance of a generic random vector $\mathbf{Z}$ is a *matrix* defined as $\mathbb{E}\left[(\mathbf{Z} - \mathbb{E}\left[\mathbf{Z}\right])(\mathbf{Z} - \mathbb{E}\left[\mathbf{Z}\right])^T\right]$. In particular the diagonal terms contains the variances of the single components of $\mathbf{Z}$, while the other elements refer to the covariances of distinct components. We have:

$$\begin{aligned}
\mathbb{V}\left[\theta^\star\right] &= \mathbb{E}\left[(\hat{\theta} - \theta^\star)(\hat{\theta} - \theta^\star)^T\right] \\
&= \mathbb{E}\left[[(\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon}][(\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon}]^T\right] && \text{(By Equation (5))} \\
&= \mathbb{E}\left[(\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\Phi(\Phi^T\Phi)^{-1}\right] && (\Phi^T\Phi \text{ is symmetric, and so its inverse}) \\
&= \mathbb{E}_{\mathbf{X}}[(\Phi^T\Phi)^{-1}\Phi^T\mathbb{E}_{\boldsymbol{\varepsilon}}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T]\Phi(\Phi^T\Phi)^{-1}] && (\mathbf{X} \text{ and } \boldsymbol{\varepsilon} \text{ are independent}) \\
&= \sigma^2\mathbb{E}_{\mathbf{X}}[(\Phi^T\Phi)^{-1}\Phi^T\Phi(\Phi^T\Phi)^{-1}] = \sigma^2\mathbb{E}\left[(\Phi^T\Phi)^{-1}\right].
\end{aligned}$$

Note, in the second to last equality, we crucially rely on the fact that $\mathbb{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\right]$ is equal to the identity matrix multiplied by σ^2. Indeed, the various terms ε_i and ε_j are independent (as they refer to distinct independent samples), while $\mathbb{E}\left[\varepsilon_i^2\right] = \sigma^2$ by definition. □

Remark 3.4 (Variance Rate)**.** *The factor $1/n$ in the definition of $\hat{\Sigma}$ seems a bit artificial. However, this is not the case: typically, by the law of large numbers, the value of $\frac{1}{n}\Phi^T\Phi$ tends to Σ, where Σ is the deterministic matrix $\mathbb{E}\left[\phi(X)\phi(X)^T\right]$ that only depends on the marginal distribution of $\mathcal{D}$ on $\mathcal{X}$. This means that the variance of the estimator decreases linearly in $1/n$.*

Important

We are trying to estimate the predictor $\phi(X)^T\theta^\star$ which minimizes the *expected* risk $\mathbb{E}\left[(Y - \phi(X)^T\theta^\star)^2\right]$, with the predictor $\phi(X)^T\hat{\theta}$, which minimizes the *empirical* mean-squared error: $\sum_{i=1}^n(\phi(x_i)^T\theta - y_i)^2$. Since we do not have access to the distribution $\mathcal{D}$, we are using the n samples as a *proxy* for it!

3.2 The Risk of the OLS Estimator

We can compute explicitly the expected risk $R(\theta)$ associated with any predictor of the form $f_\theta(x)$.

Proposition 3.5. *Under the linear assumption, for any $\theta \in \mathbb{R}^d$, we have*

$$R(\theta) = (\theta^\star - \theta)^T \Sigma (\theta^\star - \theta) + R(\hat{\theta}^\star),$$

where the matrix Σ is defined as $\mathbb{E}\left[\phi(X)\phi(X)^T\right]$. Moreover, $R(\theta^\star) = \sigma^2$.

Proof. Consider a generic vector θ, we have the following chain of inequalities:

$$\begin{aligned}
R(\theta) &= \mathbb{E}\left[(Y - \phi(X)^T\theta)^2\right] && \text{(By definition of expected risk)}\\
&= \mathbb{E}\left[(\phi(X)^T(\theta^\star - \theta) + \varepsilon)^2\right] && \text{(By the linear assumption)}\\
&= \mathbb{E}\left[(\phi(X)^T(\theta^\star - \theta))^2\right] + 2\mathbb{E}\left[\varepsilon(\phi(X)^T(\theta^\star - \theta)\right] + \mathbb{E}\left[\varepsilon^2\right]\\
&= \mathbb{E}\left[(\phi(X)^T(\theta^\star - \theta))^2\right] + 2\mathbb{E}\left[\varepsilon\right]\mathbb{E}\left[(\phi(X)^T(\theta^\star - \theta)\right] + \mathbb{E}\left[\varepsilon^2\right] && (X \text{ and } \varepsilon \text{ are independent})\\
&= \mathbb{E}\left[(\phi(X)^T(\theta^\star - \theta))^2\right] + \sigma^2 && (\text{By assumption, } \mathbb{E}\left[\varepsilon\right] = 0,\ \mathbb{E}\left[\varepsilon^2\right] = \sigma^2)\\
&= \mathbb{E}\left[((\theta^\star - \theta)^T\phi(X))(\phi(X)^T(\theta^\star - \theta))\right] + \sigma^2 && (v^Tw = w^Tv \text{ for any vectors } v \text{ and } w)\\
&= (\theta^\star - \theta)^T\mathbb{E}\left[\phi(X)\phi(X)^T\right](\theta^\star - \theta) + \sigma^2. && (\theta \text{ and } \theta^\star \text{ are deterministic})
\end{aligned}$$

Note, if $\theta = \theta^\star$, then we are left with the σ^2 term. This concludes the proof. □

Proposition 3.6. *Under the linear model, the expected risk of the OLS estimator is*

$$R(\hat{\theta}) = R(\theta) + \frac{\sigma^2}{n}\mathbb{E}\left[\Sigma\hat{\Sigma}^{-1}\right]$$

Proof. The OLS estimator is a random variable that depends on n i.i.d. samples from the linear model, we denote such samples with $(\mathbf{X}, \mathbf{Y})$, where the ith component of $\mathbf{X}$ is the random variable corresponding to the ith sample. We are in the linear model, therefore $\mathbf{Y} = \Phi(\mathbf{X})\theta^\star + \boldsymbol{\varepsilon}$, where $\boldsymbol{\varepsilon}$ is the n-dimensional vector of the noise variables. When we compute the expected risk, we are taking the expectation with respect to an additional sample (X, Y), where $Y = \phi(X)^T\theta^\star + \varepsilon$. When we compute the expected risk of $\hat{\theta}$, we then need to take the expectation with respect to both the samples $(\mathbf{X}, \mathbf{Y})$ and (X, Y).

To start, we fix $(\mathbf{X}, \mathbf{Y})$, and take the expectation with respect to (X, Y) and apply the previous proposition on $\hat{\theta}$ that is deterministically induced by the realized samples.

$$\begin{aligned}
R(\hat{\theta}) - R(\theta^\star) &= (\theta^\star - \hat{\theta})^T\Sigma(\theta^\star - \hat{\theta})\\
&= (\theta^\star - (\Phi^T\Phi)^{-1}\Phi^T\mathbf{Y})^T\Sigma(\theta^\star - (\Phi^T\Phi)^{-1}\Phi^T\mathbf{Y}) && (\text{Definition of } \hat{\theta})\\
&= (\theta^\star - (\Phi^T\Phi)^{-1}\Phi^T(\Phi\theta^\star + \boldsymbol{\varepsilon}))^T\Sigma(\theta^\star - (\Phi^T\Phi)^{-1}\Phi^T(\Phi\theta^\star + \boldsymbol{\varepsilon})) && (\text{Linear model})\\
&= ((\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon})^T\Sigma((\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon})\\
&= \mathrm{Tr}\left((\boldsymbol{\varepsilon}^T\Phi(\Phi^T\Phi)^{-1})\Sigma((\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon})\right) && (\Phi^T\Phi \text{ is symmetric and so is its inverse})\\
&= \mathrm{Tr}\left(((\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon})(\boldsymbol{\varepsilon}^T\Phi(\Phi^T\Phi)^{-1})\Sigma\right). && (\mathrm{Tr}\,(ABC) = \mathrm{Tr}\,(BCA) = \mathrm{Tr}\,(CAB))
\end{aligned}$$

At this point, we take the expectation also with respect to the samples $(\mathbf{X}, \mathbf{Y})$, we get:

$$\begin{aligned}
R(\hat{\theta}) - R(\theta^\star) &= \mathbb{E}\left[\mathrm{Tr}\left((\Phi^T\Phi)^{-1}\Phi^T\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\Phi(\Phi^T\Phi)^{-1}\Sigma\right)\right]\\
&= \mathbb{E}_{\mathbf{X}}[\mathrm{Tr}\left((\Phi^T\Phi)^{-1}\Phi^T\mathbb{E}_{\boldsymbol{\varepsilon}}[\boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon}]\Phi(\Phi^T\Phi)^{-1}\Sigma\right)] && (\mathbf{X} \text{ and } \boldsymbol{\varepsilon} \text{ are independent})\\
&= \sigma^2\mathbb{E}_{\mathbf{X}}[\mathrm{Tr}\left((\Phi^T\Phi)^{-1}\Phi^T\Phi(\Phi^T\Phi)^{-1}\Sigma\right)] && (\mathbb{E}_{\boldsymbol{\varepsilon}}[\boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon}] = \sigma^2\mathbf{I})\\
&= \sigma^2\mathbb{E}_{\mathbf{X}}[\mathrm{Tr}\left((\Phi^T\Phi)^{-1}\Sigma\right)]\\
&= \frac{\sigma^2}{n}\mathbb{E}_{\mathbf{X}}[\mathrm{Tr}(\Sigma\hat{\Sigma}^{-1})] && (\text{Because } \mathrm{Tr}\,(AB) = \mathrm{Tr}\,(BA) \text{ and definition of } \hat{\Sigma})
\end{aligned}$$

□

Before concluding, let's highlight the differences between Σ and $\hat{\Sigma}$. Σ is a deterministic matrix, as it is given by the expected value of $\phi(X)\phi(X)^T$, *with respect to a single sample* (X, Y). The matrix $\hat{\Sigma}$ is instead a random variable, as it depends on the n i.i.d. samples represented in the vectors $(\mathbf{X}, \mathbf{Y})$, in particular, $\hat{\Sigma} = \frac{1}{n}\Phi(\mathbf{X})\Phi(\mathbf{X})^T$

4 Pointers

These notes are mainly based on Chapter 3 of Bach [2024].

References

Francis Bach. *Learning Theory from First Principles*. The MIT Press, 2024. URL `https://www.di.ens.fr/~fbach/ltfp_book.pdf`.